

LAPORAN PROJECT DATAWAREHOUSE UJIAN AKHIR SEMESTER 4 STUDI KASUS AI_JOB DATA KARYAWAN



Disusun oleh:

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**SISTEM INFORMASI BISNIS
POLITEKNIK NEGERI MALANG
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Studi Kasus Pengelolaan Data

Glolab AI Job Market & Salary Trends 2025

<https://www.kaggle.com/datasets/bismasajjad/global-ai-job-market-and-salary-trends-2025>

Menganalisis Rata-Rata Gaji di Sektor AI: Peran Jabatan, Pengalaman, Lokasi, dan Perusahaan

Dataset ini merupakan kumpulan data komprehensif yang merangkum tren pekerjaan dan gaji di bidang Kecerdasan Buatan (AI) secara global. Data ini mencakup ribuan catatan pekerjaan dari tahun 2020 hingga 2025, memberikan gambaran mendalam tentang bagaimana pasar kerja AI telah berkembang.

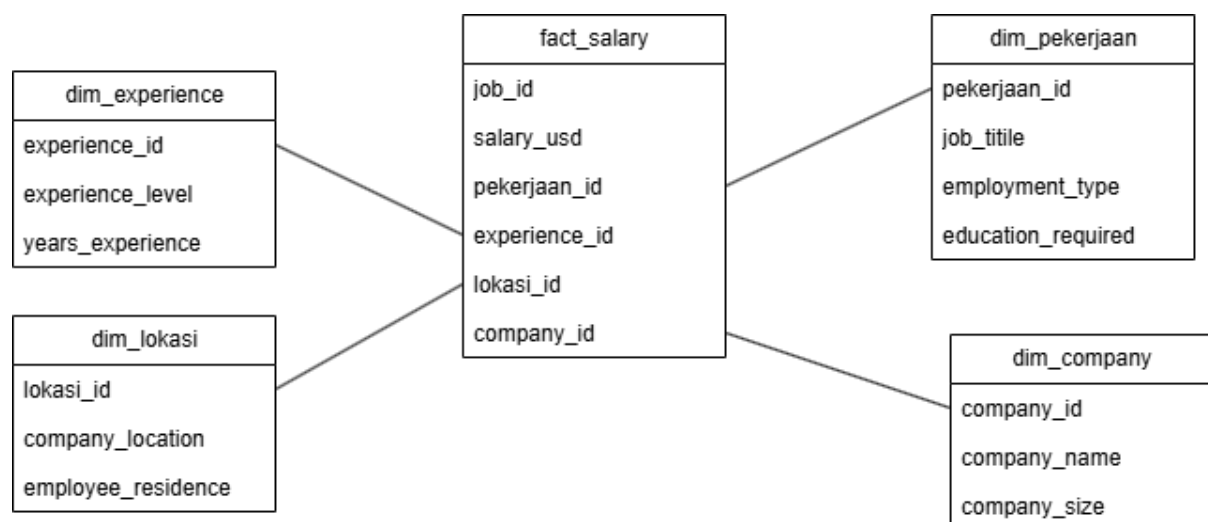
Rancangan Skema Bintang (Star Schema) untuk Data Warehouse

Tabel Fakta: fact_salart

(job_id,salary_usd,pekerjaan_id,experience_id,lokasi_id,company_id)

Tabel Dimensi:

- pekerjaan (berdasarkan job_title).
- experience (berdasarkan experience_level, years_experience).
- lokasi (berdasarkan company_location, employee_residence).
- company (berdasarkan company_name, company_size).



Langkah-Langkah Proses ETL

Langkah awal dari pembuatan Star Schema Datawarehouse ini adalah membuat database dan table lalu membuat dimensi nya terlebih dahulu dan terakhir membuat table fakta nya :

Query_dw_ai_job	<pre>-- Tabel Dimensi: Dim_Pekerjaan CREATE TABLE Dim_Pekerjaan (pekerjaan_id INT PRIMARY KEY, job_title VARCHAR(100), employment_type VARCHAR(10), education_required VARCHAR(50)) ENGINE=InnoDB; -- Tabel Dimensi: Dim_Experience CREATE TABLE Dim_Experience (experience_id INT PRIMARY KEY, experience_level VARCHAR(10), years_experience INT) ENGINE=InnoDB; -- Tabel Dimensi: Dim_Lokasi CREATE TABLE Dim_Lokasi (lokasi_id INT PRIMARY KEY, company_location VARCHAR(100), employee_residence VARCHAR(100), country_code VARCHAR(10)) ENGINE=InnoDB; -- Tabel Dimensi: Dim_Company CREATE TABLE Dim_Company (company_id INT PRIMARY KEY, company_name VARCHAR(100), company_size VARCHAR(10)) ENGINE=InnoDB; -- Tabel Fakta: fact_salary CREATE TABLE Fact_Jobs (job_id VARCHAR(10) PRIMARY KEY, salary_usd DECIMAL(15,2), pekerjaan_id INT, experience_id INT, lokasi_id INT, company_id INT, FOREIGN KEY (pekerjaan_id) REFERENCES Dim_Pekerjaan(pekerjaan_id), FOREIGN KEY (experience_id) REFERENCES Dim_Experience(experience_id), FOREIGN KEY (lokasi_id) REFERENCES Dim_Lokasi(lokasi_id), FOREIGN KEY (company_id) REFERENCES Dim_Company(company_id)) ENGINE=InnoDB;</pre>
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1. dim_pekerjaan

Rancangan Detail dim_pekerjaan.ktr :

a.Extract

(Mengambil data dari ai_job_dataset.csv yang relevan untuk Dim_Pekerjaan.)

-CSV File Input

b.Transform

(Membersihkan data dan menyiapkan untuk tabel Dim_Pekerjaan.)

-Unique Rows

-Add Sequence

c.Load

(Memuat data ke tabel Dim_Pekerjaan di data warehouse.)

-Table Output



Step name: CSV file input

Filename: D:\UAS_DATAWAREHOUSE\ai_job_dataset.csv

Delimiter: ,

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Tri
1	job_title	String		27		IDR	.	,	no
2	employment_type	String		2		IDR	.	,	no
3	education_required	String		9		IDR	.	,	no

Buttons: Help, OK, Get Fields, Preview, Cancel

Add sequence

Step name: Add sequence

Name of value: pekerjaan_id

Use a database to generate the sequence

Use DB to get sequence? ☐

Connection: dw_job_ai_2 Edit... New... Wizard...

Schema name: Schemas...

Sequence name: SEQ_ Sequences...

Use a transformation counter to generate the sequence

Use counter to calculate sequence? ☒

Counter name (optional):

Start at value: 1

Increment by: 1

Maximum value: 999999999

Help OK Cancel

Table output

Step name: table output

Connection: dw_job_ai_2 Edit... New... Wizard...

Target schema: dw_job_ai_2 Browse...

Target table: dim_pekerjaan Browse...

Commit size: 1000

Truncate table: ☒

Ignore insert errors: ☐

Specify database fields: ☒

Main options Database fields

Partition data over tables: ☐

Partitioning field:

Partition data per month: ☒

Partition data per day: ☐

Use batch update for inserts: ☒

Is the name of the table defined in a field? ☐

Field that contains name of table:

Store the tablename field: ☐

Return auto-generated key: ☐

Name of auto-generated key field:

Help OK Cancel SQL

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

2025/06/15 15:26:22 - Spoon - Transformation opened.

2025/06/15 15:26:22 - Spoon - Launching transformation [dim_experience]...

2025/06/15 15:26:22 - Spoon - Started the transformation execution.

2025/06/15 15:26:22 - dim_experience - Dispatching started for transformation [dim_experience]

2025/06/15 15:26:22 - Table output.0 - Connected to database [dw_job_ai_2] (commit=1000)

2025/06/15 15:26:22 - CSV file input.0 - Header row skipped in file 'D:/UAS_DATAWAREHOUSE/ai_job_dataset.csv'

2025/06/15 15:26:22 - CSV file input.0 - Finished processing (I=15001, O=0, R=0, W=15000, U=0, E=0)

2025/06/15 15:26:22 - Unique rows.0 - Finished processing (I=0, O=0, R=15000, W=13969, U=0, E=0)

2025/06/15 15:26:23 - Add sequence.0 - Finished processing (I=0, O=0, R=13969, W=13969, U=0, E=0)

2025/06/15 15:26:24 - Table output.0 - Finished processing (I=0, O=0, R=13969, W=13969, U=0, E=0)

2025/06/15 15:26:24 - Spoon - The transformation has finished!!

2025/06/15 15:28:09 - Spoon - Transformation opened.

2025/06/15 15:28:09 - Spoon - Launching transformation [fact_jobs_salary]...

2025/06/15 15:28:09 - Spoon - Started the transformation execution.

2025/06/15 15:28:14 - Spoon - The transformation has finished!!

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

First rows Last rows Off

#	experience_level	years_experience
1	SE	9
2	EN	1
3	MI	2
4	SE	7
5	EN	0
6	SE	7
7	MI	3
8	EN	0
9	SE	7
1.	SE	5
1.	SE	8
1.	EX	15
1.	SE	5
1.	EN	0
1.	SE	6
1.	EN	0
1.	EN	1
1.	EX	15
1.	SE	9

2. dim_experience

Rancangan Detail dim_experience.ktr :

a.Extract

(Mengambil data dari ai_job_dataset.csv yang relevan untuk dim_experience.)

-CSV File Input

b.Transform

(Membersihkan data dan menyiapkan untuk tabel dim_experience.)

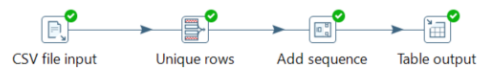
-Unique Rows

-Add Sequence

c.Load

(Memuat data ke tabel dim_experience di data warehouse.)

-Table Output



CSV file input

Step name: CSV file input

Filename: D:/UAS_DATAWAREHOUSE/ai_job_dataset.csv

Delimiter: ,

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result? ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim
1	experience_level	String		2		IDR	.	,	none
2	years_experience	Integer	#	15	0	IDR	.	,	none

Help OK Get Fields Preview Cancel

Unique rows

Step name: Unique rows

Settings

Add counter to output? ☐ Counter field:

Redirect duplicate row ☐ Error description:

Fields to compare on (no entries means: compare complete row)

#	Fieldname	Ignore case
1	experience_level	N
2	years_experience	N

Help OK Cancel Get

Add sequence

Step name: Add sequence

Name of value: experience_id

Use a database to generate the sequence

Use DB to get sequence? ☐

Connection: dw_job_ai_2 Edit... New... Wizard...

Schema name: Schemas...

Sequence name: SEQ_ Sequences...

Use a transformation counter to generate the sequence

Use counter to calculate sequence? ☒

Counter name (optional):

Start at value: 1

Increment by: 1

Maximum value: 999999999

Help OK Cancel

Table output

Step name: Table output

Connection: dw_job_ai_2 Edit... New... Wizard...

Target schema: dw_job_ai_2 Browse...

Target table: dim_experience Browse...

Commit size: 1000

Truncate table: ☒

Ignore insert errors: ☐

Specify database fields: ☒

Main options Database fields

Partition data over tables: ☐

Partitioning field:

Partition data per month: ☐

Partition data per day: ☐

Use batch update for inserts: ☒

Is the name of the table defined in a field? ☐

Field that contains name of table:

Store the tablename field: ☐

Return auto-generated key: ☐

Name of auto-generated key field:

Help OK Cancel SQL

2025/06/15 15:26:22 - Spoon - Transformation opened.

2025/06/15 15:26:22 - Spoon - Launching transformation [dim_experience]...

2025/06/15 15:26:22 - Spoon - Started the transformation execution.

2025/06/15 15:26:22 - dim_experience - Dispatching started for transformation [dim_experience]

2025/06/15 15:26:22 - Table output.0 - Connected to database [dw_job_ai_2] (commit=1000)

2025/06/15 15:26:22 - CSV file input.0 - Header row skipped in file 'D:/UAS_DATAWAREHOUSE/ai_job_dataset.csv'

2025/06/15 15:26:22 - CSV file input.0 - Finished processing (I=15001, O=0, R=0, W=15000, U=0, E=0)

2025/06/15 15:26:22 - Unique rows.0 - Finished processing (I=0, O=0, R=15000, W=13969, U=0, E=0)

2025/06/15 15:26:23 - Add sequence.0 - Finished processing (I=0, O=0, R=13969, W=13969, U=0, E=0)

2025/06/15 15:26:24 - Table output.0 - Finished processing (I=0, O=13969, R=13969, W=13969, U=0, E=0)

2025/06/15 15:26:24 - Spoon - The transformation has finished!!

2025/06/15 15:28:09 - Spoon - Transformation opened.

2025/06/15 15:28:09 - Spoon - Launching transformation [fact_jobs_salary]...

2025/06/15 15:28:09 - Spoon - Started the transformation execution.

2025/06/15 15:28:14 - Spoon - The transformation has finished!!

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

First rows Last rows Off

#	experience_level	years_experience	experience_id
1	SE	9	1
2	EN	1	2
3	MI	2	3
4	SE	7	4
5	EN	0	5
6	SE	7	6
7	MI	3	7
8	EN	0	8
9	SE	7	9
1.	SE	5	10
1.	SE	8	11
1.	EX	15	12
1.	SE	5	13
1.	EN	0	14
1.	SE	6	15
1.	EN	0	16
1.	EN	1	17
1.	EX	15	18
1.	SE	9	19
2.	MI	4	20
2.	MI	3	21
2.	MI	2	22
2.	SE	7	23
2.	MI	4	24
2.	MI	2	25

3. dim_company :

Rancangan Detail dim_company.ktr :

1.Extract

(Mengambil data dari ai_job_dataset.csv yang relevan untuk Dim_company.)

-CSV File Input

2.Transform

(Membersihkan data dan menyiapkan untuk tabel Dim_company.)

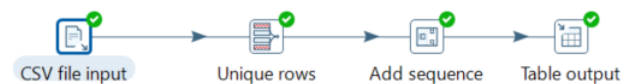
-Unique Rows

-Add Sequence

3.Load

(Memuat data ke tabel Dim_company di data warehouse.)

-Table Output



CSV file input

Step name: CSV file input

Filename: D:/UAS_DATAWAREHOUSE/ai_job_dataset.csv

Delimiter: ,

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result? ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim t
1	company_name	String		26		IDR	.	,	none
2	company_size	String		1		IDR	.	,	none

Help OK Get Fields Preview Cancel

Unique rows

Step name: Unique rows

Settings

Add counter to output? ☐ Counter field:

Redirect duplicate row ☐ Error description:

Fields to compare on (no entries means: compare complete row)

#	Fieldname	Ignore case
1	company_name	N
2	company_size	N

Help OK Cancel Get

Add sequence

Step name

Add sequence

Name of value

company_id

Use a database to generate the sequence

Use DB to get sequence?

☐

Connection

dw_job_ai_2

Edit...

New...

Wizard...

Schema name

Schemas...

Sequence name

SEQ_

Sequences...

Use a transformation counter to generate the sequence

Use counter to calculate sequence?

☒

Counter name (optional)

Start at value

1

Increment by

1

Maximum value

999999999

Help

OK

Cancel

Table output

Step name

Table output

Connection

dw_job_ai_2

Edit...

New...

Wizard...

Target schema

dw_job_ai_2

Browse...

Target table

dim_company

Browse...

Commit size

1000

Truncate table

☒

Ignore insert errors

☐

Specify database fields

☒

Main options

Database fields

Partition data over tables

☐

Partitioning field

Partition data per month

☒

Partition data per day

☐

Use batch update for inserts

☒

Is the name of the table defined in a field?

☐

Field that contains name of table:

Store the tablename field

☐

Return auto-generated key

☐

Name of auto-generated key field

Help

OK

Cancel

SQL

Execution Results

Logging

Execution History

Step Metrics

Performance Graph

Metrics

Preview data

2025/06/15 15:27:12 - Spoon - Transformation opened.

2025/06/15 15:27:12 - Spoon - Launching transformation [dim_company]...

2025/06/15 15:27:12 - Spoon - Started the transformation execution.

2025/06/15 15:27:12 - dim_company - Dispatching started for transformation [dim_company]

2025/06/15 15:27:12 - Table output.0 - Connected to database [dw_job_ai_2] (commit=1000)

2025/06/15 15:27:12 - CSV file input.0 - Header row skipped in file 'D:/UAS_DATAWAREHOUSE/ai_job_dataset.csv'

2025/06/15 15:27:12 - CSV file input.0 - Finished processing (I=15001, O=0, R=0, W=15000, U=0, E=0)

2025/06/15 15:27:12 - Unique rows.0 - Finished processing (I=0, O=0, R=15000, W=14681, U=0, E=0)

2025/06/15 15:27:13 - Add sequence.0 - Finished processing (I=0, O=0, R=14681, W=14681, U=0, E=0)

2025/06/15 15:27:14 - Table output.0 - Finished processing (I=0, O=0, R=14681, W=14681, U=0, E=0)

2025/06/15 15:27:14 - Spoon - The transformation has finished!!

2025/06/15 15:28:09 - Spoon - Transformation opened.

2025/06/15 15:28:09 - Spoon - Launching transformation [fact_jobs_salary]...

2025/06/15 15:28:09 - Spoon - Started the transformation execution.

2025/06/15 15:28:14 - Spoon - The transformation has finished!!

Execution Results

Logging

Execution History

Step Metrics

Performance Graph

Metrics

Preview data

First rows

Last rows

Off

#	company_name	company_size	company_id
1	Smart Analytics	M	1
2	TechCorp Inc	M	2
3	Autonomous Tech	L	3
4	Future Systems	M	4
5	Advanced Robotics	S	5
6	Neural Networks Co	M	6
7	DataVision Ltd	S	7
8	Cloud AI Solutions	L	8
9	Quantum Computing Inc	L	9
1.	Cloud AI Solutions	M	10
1.	Smart Analytics	M	11
1.	TechCorp Inc	S	12
1.	Predictive Systems	M	13
1.	Cloud AI Solutions	S	14
1.	Advanced Robotics	M	15
1.	DataVision Ltd	L	16
1.	Predictive Systems	L	17
1.	AI Innovations	L	18
1.	Algorithmic Solutions	M	19
2.	AI Innovations	L	20
2.	AI Innovations	M	21
2.	Cognitive Computing	M	22
2.	Quantum Computing Inc	S	23
2.	Cloud AI Solutions	M	24
2.	AI Innovations	S	25
2.	DataVision Ltd	L	26

4. dim_lokasi

Rancangan Detail dim_lokasi.ktr :

1.Extract

(Mengambil data dari ai_job_dataset.csv yang relevan untuk dim_lokasi.)

-CSV File Input

2.Transform

(Membersihkan data dan menyiapkan untuk tabel dim_lokasi.)

-Unique Rows

-Add Sequence

3.Load

(Memuat data ke tabel dim_lokasi di data warehouse.)

-Table Output



CSV file input

Step name: CSV file input

Filename: D:/UAS_DATAWAREHOUSE/ai_job_dataset.csv

Delimiter: ,

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result? ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Ti
1	company_location	String		14		IDR	.	,	n
2	employee_residence	String		14		IDR	.	,	n

Help OK Get Fields Preview Cancel

Unique rows

Step name: Unique rows

Settings

Add counter to output? ☐ Counter field

Redirect duplicate row ☐ Error description

Fields to compare on (no entries means: compare complete row)

#	Fieldname	Ignore case
1	company_location	N
2	employee_residence	N

Help OK Cancel Get

Add sequence

Step name: Add sequence

Name of value: lokasi_id

Use a database to generate the sequence

Use DB to get sequence? ☐

Connection: dw_job_ai_2 Edit... New... Wizard...

Schema name: SEQ_ Schemas...

Sequence name: SEQ_ Sequences...

Use a transformation counter to generate the sequence

Use counter to calculate sequence? ☒

Counter name (optional):

Start at value: 1

Increment by: 1

Maximum value: 999999999

Help OK Cancel

Table output

Step name: table output

Connection: dw_job_ai_2 Edit... New... Wizard...

Target schema: dw_job_ai_2 Browse...

Target table: dim_lokasi Browse...

Commit size: 1000

Truncate table: ☒

Ignore insert errors: ☐

Specify database fields: ☒

Main options Database fields

Partition data over tables: ☐

Partitioning field:

Partition data per month: ☐

Partition data per day: ☐

Use batch update for inserts: ☒

Is the name of the table defined in a field? ☐

Field that contains name of table:

Store the tablename field: ☐

Return auto-generated key: ☐

Name of auto-generated key field:

Help OK Cancel SQL

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

2025/06/15 15:27:30 - Spoon - Transformation opened.

2025/06/15 15:27:30 - Spoon - Launching transformation [dim_lokasi]...

2025/06/15 15:27:30 - Spoon - Started the transformation execution.

2025/06/15 15:27:30 - dim_lokasi - Dispatching started for transformation [dim_lokasi]

2025/06/15 15:27:30 - Table output.0 - Connected to database [dw_job_ai_2] (commit=1000)

2025/06/15 15:27:30 - CSV file input.0 - Header row skipped in file 'D:/UAS_DATAWAREHOUSE/ai_job_dataset.csv'

2025/06/15 15:27:30 - CSV file input.0 - Finished processing (I=15001, O=0, R=0, W=15000, U=0, E=0)

2025/06/15 15:27:30 - Unique rows.0 - Finished processing (I=0, O=0, R=15000, W=14614, U=0, E=0)

2025/06/15 15:27:31 - Add sequence.0 - Finished processing (I=0, O=0, R=14614, W=14614, U=0, E=0)

2025/06/15 15:27:32 - Table output.0 - Finished processing (I=0, O=14614, R=14614, W=14614, U=0, E=0)

2025/06/15 15:27:32 - Spoon - The transformation has finished!!

2025/06/15 15:28:09 - Spoon - Transformation opened.

2025/06/15 15:28:09 - Spoon - Launching transformation [fact_jobs_salary]...

2025/06/15 15:28:09 - Spoon - Started the transformation execution.

2025/06/15 15:28:14 - Spoon - The transformation has finished!!

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

First rows Last rows Off

#	company_location	employee_residence	lokasi_id
1	China	China	1
2	Canada	Ireland	2
3	Switzerland	South Korea	3
4	India	India	4
5	France	Singapore	5
6	Germany	Germany	6
7	United Kingdom	United Kingdom	7
8	France	France	8
9	Singapore	Singapore	9
10	Austria	Austria	10
11	Sweden	Sweden	11
12	South Korea	South Korea	12
13	France	France	13
14	Norway	Norway	14
15	Sweden	Sweden	15
16	United Kingdom	Israel	16
17	India	India	17
18	France	France	18
19	Singapore	Singapore	19
20	Netherlands	Germany	20
21	Norway	France	21
22	United States	United States	22

5. fact_salary

Rancangan Detail fact_salary.ktr :

1.Extract

(Mengambil data dari ai_job_dataset.csv yang relevan untuk dim_lokasi.)

-CSV File Input

2.Transform

(Membersihkan data dan menyiapkan untuk tabel dim_lokasi.)

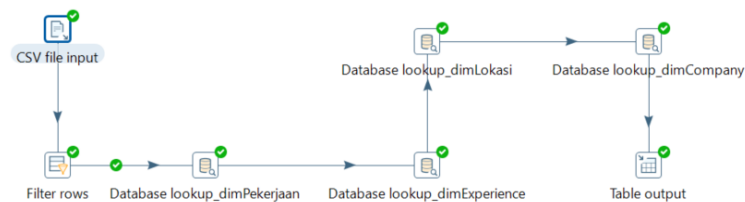
-Filter Rows

-Database Lookup

3.Load

(Memuat data ke tabel dim_lokasi di data warehouse.)

-Table Output



CSV file input

Step name: CSV file input

Filename: D:/UAS_DATAWAREHOUSE/ai_job_dataset.csv

Delimiter: ,

Enclosure: "

NIO buffer size: 50000

Lazy conversion? ☒

Header row present? ☒

Add filename to result: ☐

The row number field name (optional):

Running in parallel? ☐

New line possible in fields? ☐

Format: mixed

File encoding:

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Tr
1	job_id	String		7		IDR	.	,	n
2	salary_usd	Integer	#	15	0	IDR	.	,	n
3	job_title	String		27		IDR	.	,	n
4	experience_level	String		2		IDR	.	,	n
5	years_experience	Integer	#	15	0	IDR	.	,	n
6	company_location	String		14		IDR	.	,	n
7	employee_residence	String		14		IDR	.	,	n
8	company_name	String		26		IDR	.	,	n
9	company_size	String		1		IDR	.	,	n

Help OK Get Fields Preview Cancel

Filter rows

Step name: Filter rows

Send 'true' data to step: Database lookup_dimPekerjaan

Send 'false' data to step:

The condition:

☐ salary_usd >= [0]

AND

salary_usd IS NOT NULL

Help OK Cancel

Database lookup

Step name

Database lookup_dimPekerjaan

Connection

dw_job_ai_2

Edit...

New...

Wizard...

Lookup schema

dw_job_ai_2

Browse...

Lookup table

dim_pekerjaan

Browse...

Enable cache?

☐

Cache size in rows (0=cache)

0

Load all data from table

☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	job_title	=	job_title	

Values to return from the lookup table :

#	Field	New name	Default	Type
1	pekerjaan_id		-1	Integer

Do not pass the row if the lookup fails

☐

Fail on multiple results?

☐

Order by

Help

OK

Cancel

Get Fields

Get lookup fields

Database lookup

Step name

Database lookup_dimExperience

Connection

dw_job_ai_2

Edit...

New...

Wizard...

Lookup schema

dw_job_ai_2

Browse...

Lookup table

dim_experience

Browse...

Enable cache?

☐

Cache size in rows (0=cache)

0

Load all data from table

☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	experience_level	=	experience_level	
2	years_experience	=	years_experience	

Values to return from the lookup table :

#	Field	New name	Default	Type
1	experience_id		-1	Integer

Do not pass the row if the lookup fails

☐

Fail on multiple results?

☐

Order by

Help

OK

Cancel

Get Fields

Get lookup fields

Database lookup

Step name: Database lookup_dimLokasi

Connection: dw_job_ai_2

Lookup schema: dw_job_ai_2

Lookup table: dim_lokasi

Enable cache? ☐

Cache size in rows (0=cache): 0

Load all data from table ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	company_location	=	company_location	
2	employee_residence	=	employee_residence	

Values to return from the lookup table:

#	Field	New name	Default	Type
1	lokasi_id		-1	Integer

Do not pass the row if the lookup fails ☐

Fail on multiple results? ☐

Order by:

Help OK Cancel Get Fields Get lookup fields

Database lookup

Step name: Database lookup_dimCompany

Connection: dw_job_ai_2

Lookup schema: dw_job_ai_2

Lookup table: dim_company

Enable cache? ☐

Cache size in rows (0=cache): 0

Load all data from table ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	company_name	=	company_name	
2	company_size	=	company_size	

Values to return from the lookup table:

#	Field	New name	Default	Type
1	company_id		-1	Integer

Do not pass the row if the lookup fails ☐

Fail on multiple results? ☐

Order by:

Help OK Cancel Get Fields Get lookup fields

Table output

Step name: Table output

Connection: dw_job_ai_2

Target schema: dw_job_ai_2

Target table: fact_salary

Commit size: 1000

Truncate table ☒

Ignore insert errors ☐

Specify database fields ☒

Main options Database fields

Partition data over tables ☐

Partitioning field:

Partition data per month ☒

Partition data per day ☐

Use batch update for inserts ☒

Is the name of the table defined in a field? ☐

Field that contains name of table:

Store the tablename field ☒

Return auto-generated key ☐

Name of auto-generated key field:

Help OK Cancel SQL

Execution Results

Logging

Execution History

Step Metrics

Performance Graph

Metrics

Preview data

2025/06/15 15:28:09 - Spoon - Transformation opened.

2025/06/15 15:28:09 - Spoon - Launching transformation [fact_jobs_salary]...

2025/06/15 15:28:09 - Spoon - Started the transformation execution.

2025/06/15 15:28:09 - fact_jobs_salary - Dispatching started for transformation [fact_jobs_salary]

2025/06/15 15:28:09 - Table output.0 - Connected to database [dw_job_ai_2] (commit=1000)

2025/06/15 15:28:09 - CSV file input.0 - Header row skipped in file 'D:/UAS_DATAWAREHOUSE/ai_job_dataset.csv'

2025/06/15 15:28:09 - CSV file input.0 - Finished processing (I=15001, O=0, R=0, W=15000, U=0, E=0)

2025/06/15 15:28:10 - Filter rows.0 - Finished processing (I=0, O=0, R=15000, W=15000, U=0, E=0)

2025/06/15 15:28:12 - Database lookup_dimPekerjaan.0 - Finished processing (I=15000, O=0, R=15000, W=15000, U=0, E=0)

2025/06/15 15:28:12 - Database lookup_dimExperience.0 - Finished processing (I=15000, O=0, R=15000, W=15000, U=0, E=0)

2025/06/15 15:28:14 - Database lookup_dimLokasi.0 - Finished processing (I=15000, O=0, R=15000, W=15000, U=0, E=0)

2025/06/15 15:28:14 - Database lookup_dimCompany.0 - Finished processing (I=15000, O=0, R=15000, W=15000, U=0, E=0)

2025/06/15 15:28:14 - Table output.0 - Finished processing (I=0, O=15000, R=15000, W=15000, U=0, E=0)

2025/06/15 15:28:14 - Spoon - The transformation has finished!!

Execution Results

Logging

Execution History

Step Metrics

Performance Graph

Metrics

Preview data

First rows

Last rows

Off

#	job_id	salary_usd	job_title	experience_level	years_experience	company_location	employee_residence	company_name	company_size	pekerjaan_id	experience_id	lokasi_id	company_id
1	AI00001	90376	AI Research Scientist	SE	9	China	China	Smart Analytics	M	1	1	1	1
2	AI00002	61895	AI Software Engineer	EN	1	Canada	Ireland	TechCorp Inc.	M	2	2	2	2
3	AI00003	152626	AI Specialist	MI	2	Switzerland	South Korea	Autonomous Tech	L	3	3	3	3
4	AI00004	80215	NLP Engineer	SE	7	India	India	Future Systems	M	4	4	4	4
5	AI00005	54624	AI Consultant	EN	0	France	Singapore	Advanced Robotics	S	5	5	5	5
6	AI00006	123574	AI Architect	SE	7	Germany	Germany	Neural Networks Co	M	6	4	6	6
7	AI00007	79670	Principal Data Scientist	MI	3	United Kingdom	United Kingdom	DataVision Ltd	S	7	7	7	7
8	AI00008	70640	NLP Engineer	EN	0	France	France	Cloud AI Solutions	L	4	5	8	8
9	AI00009	160710	Data Analyst	SE	7	Singapore	Singapore	Quantum Computing Inc.	L	9	4	9	9
10	AI00010	102557	AI Software Engineer	SE	5	Austria	Austria	Cloud AI Solutions	M	2	10	10	10

Hasil query atau Dashboard.

```
SELECT
    p.job_title,
    e.experience_level,
    l.company_location,
    c.company_name,
    AVG(f.salary_usd) as average_salary
FROM fact_salary f
JOIN Dim_Pekerjaan p ON f.pekerjaan_id = p.pekerjaan_id
JOIN Dim_Experience e ON f.experience_id = e.experience_id
JOIN Dim_Lokasi l ON f.lokasi_id = l.lokasi_id
JOIN Dim_Company c ON f.company_id = c.company_id
GROUP BY p.job_title, e.experience_level, l.company_location,
c.company_name
HAVING AVG(f.salary_usd) > 50000
ORDER BY average_salary DESC;
```

Hasil Menampilkan rata-rata gaji untuk setiap kombinasi pekerjaan, level pengalaman, lokasi perusahaan, dan nama perusahaan, dengan filter gaji di atas \$50,000.

job_title	experience_level	company_location	company_name	average_salary
AI Architect	EX	Switzerland	Future Systems	390613.000000
Data Scientist	EX	Switzerland	DeepTech Ventures	388754.000000
AI Software Engineer	EX	Denmark	Algorithmic Solutions	379418.000000
AI Specialist	EX	Switzerland	DataVision Ltd	373526.000000
AI Specialist	EX	Denmark	Machine Intelligence Group	370038.000000
AI Product Manager	EX	Switzerland	Neural Networks Co	367141.000000
Computer Vision Engineer	EX	Switzerland	Machine Intelligence Group	366957.000000
Principal Data Scientist	EX	Switzerland	Digital Transformation LLC	364635.000000
Deep Learning Engineer	EX	Norway	DeepTech Ventures	362503.000000
Data Scientist	EX	Norway	Advanced Robotics	361723.000000
Head of AI	EX	Denmark	Quantum Computing Inc	361718.000000
Data Analyst	EX	Denmark	Algorithmic Solutions	361541.000000
AI Specialist	EX	Denmark	Algorithmic Solutions	358673.000000
NLP Engineer	EX	Switzerland	Machine Intelligence Group	357880.000000

Analisis KPI (Key Performance Indicators)

Waktum Proses ETL :

Berdasarkan pengalaman tipikal dengan dataset kecil seperti ai_job_dataset.csv (misalnya, ratusan baris), waktu proses ETL untuk setiap .ktr biasanya berkisar antara 5-30 detik tergantung pada ukuran data, kecepatan koneksi database, dan konfigurasi hardware.

- **Ekstraksi:** Membaca file CSV cepat (<5 detik).
- **Transformasi:** Mapping dan lookup (misalnya, **Database Lookup**) mungkin memakan 10-20 detik jika ada banyak baris atau koneksi lambat.
- **Pemuatan:** Menulis ke database (terutama dengan foreign key checks) bisa memakan 5-10 detik.

Optimasi: Menggunakan cache di **Database Lookup** dan menjalankan dalam batch dapat mengurangi waktu. Untuk estimasi akurat, lihat tab **Performance Graph** di **Execution Results** setelah menjalankan .ktr.

Konsistensi dan Kualitas Data:

- **Konsistensi:** Data di warehouse sekarang konsisten karena menggunakan model Star Schema dengan foreign key yang terdefinisi. Namun, jika .ktr dimensi dijalankan ulang tanpa kontrol (seperti yang menyebabkan error sebelumnya), konsistensi bisa terganggu. Dengan pendekatan "hapus dan mulai ulang" yang disarankan, konsistensi terjaga selama semua .ktr dijalankan dalam urutan yang benar.

- **Kualitas:**

- Data telah dibersihkan (misalnya, null di salary_usd diganti 0 via **If Null Value**).
- Duplikat dihapus dengan **Unique Rows** di dimensi.
- Namun, kualitas tergantung pada data sumber (ai_job_dataset.csv). Jika ada data tidak valid (misalnya, salary_usd negatif sebelum filter), itu akan dihapus, yang mengurangi jumlah baris tapi meningkatkan akurasi.

Akurasi dan Kelengkapan Data dalam Warehouse:

Akurasi: Data di warehouse akurat sejauh data sumbernya benar. Proses ETL telah memfilter data invalid (misalnya, salary_usd <= 0 dihapus), dan lookup ke dimensi memastikan foreign key sesuai. Namun, jika ada kesalahan input (misalnya, job_title tidak cocok di Dim_Pekerjaan), akan ada baris dengan nilai default (misalnya, -1), yang perlu diperiksa.